



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 11

Assignment: Refining the Object-Oriented Design

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

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ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 11 module and your lecture notes before attempting each question.
- Use the university course registration case study for your examples.
- UML notation must be correct — visibility (-, +), types, parameters and return values.
- Where you critique a design, propose a concrete improvement, not just a description.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define design refinement and explain its purpose. [2 marks]

Question A2

Differentiate an analysis class from a design class, with an example. [3 marks]

Question A3

Define entity, boundary and control classes, giving one example of each. [3 marks]

Question A4

Explain the purpose of adding visibility to attributes and operations. [2 marks]

Question A5

What is a parameter? Differentiate it from an attribute. [2 marks]

Question A6

Define a return type and explain why it matters in design. [2 marks]

Question A7

Explain the relationship between analysis and design models. [2 marks]

Question A8

What is traceability and why is it important in refinement? [2 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

Consider the analysis class Student with attributes and operations.

- (a) Refine Student into a design class with typed attributes. [3 marks]
- (b) Add typed parameters and return types to operations. [3 marks]
- (c) Add visibility markers to attributes and operations. [3 marks]

- (d) Explain your visibility decisions. [3 marks]

Question B2 (8 marks)

Consider the analysis relationship Student — registers for — Course.

- (a) Draw this as a design class diagram with visibility. [4 marks]
- (b) Add an association class Registration if appropriate. [4 marks]

Section C — Case Study (10 marks)

Consider the university course registration system with Student, Course, Lecturer and Registration.

As an object-oriented designer:

- (a) Refine the analysis model into a design model with typed attributes and operations. [4 marks]
- (b) Add visibility and explain your decisions. [3 marks]
- (c) Show how the design model maintains traceability to the analysis model. [3 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (8 questions)	20
B	Application questions (12 + 8)	20
C	Case study — registration system	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Where you refine models, show the before and after versions clearly.
- Cite any sources you consult using the format used in the Week 11 module references.